Experiment No. 01: Classify various types of cabling used in networking.

LEARNING OBJECTIVE: Students should be able to study different types of network cables and practically implement the cross-wired cable and straight through cable using clamping tool.

TOOLS: and Google Docs

THEORY:

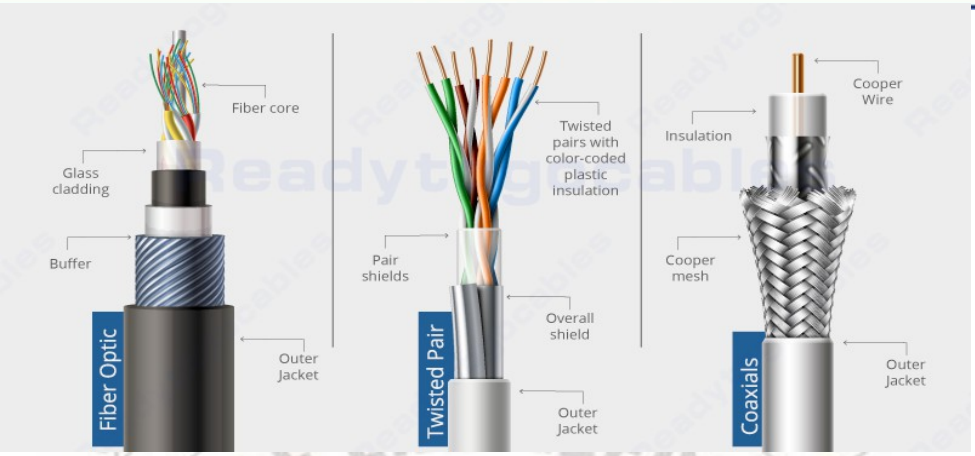
Network cables form the physical backbone of computer networks, acting as the medium for transmitting data between devices such as computers, switches, and routers. Choosing the correct type of cable and wiring method is crucial to ensure reliable and efficient communication. Among the various networking media, twisted pair cables (specifically Unshielded Twisted Pair – UTP) are widely used in LAN (Local Area Network) environments due to their low cost, flexibility, and ease of installation. Two primary wiring standards, straight-through and cross-wired (crossover), are implemented depending on the type of devices being connected.

Types of Network Cables

1. **Twisted Pair Cable (UTP and STP):**
   - **UTP (Unshielded Twisted Pair):** Common in Ethernet LANs; consists of 4 pairs of twisted copper wires. Twisting reduces electromagnetic interference (EMI).
   - **STP (Shielded Twisted Pair):** Similar to UTP but includes shielding to provide extra protection against EMI, used in environments with high interference.

2. **Coaxial Cable:**
   - Used in older Ethernet systems and cable television. It has a central conductor, insulating layer, metallic shield, and protective outer layer. Offers better resistance to noise but is less flexible compared to UTP.

3. **Fiber Optic Cable:**
   - Uses light signals to transmit data, offering extremely high bandwidth and long-distance communication with immunity to EMI. Commonly used in backbone networks, data centers, and long-distance internet connections.



Straight-Through Cable

- **Definition:** A straight-through cable is a type of Ethernet cable where the pin configuration on both ends of the cable is identical, typically following either the **TIA/EIA-568A** or **TIA/EIA-568B** wiring standard.
- **Purpose:** Used to connect different types of devices (e.g., PC to switch, switch to router).
- **Wiring:** Both ends follow the same color code sequence. Example: If one end follows 568B, the other must also follow 568B.

Applications:

- Connecting a computer to a switch/hub.
- Connecting a router to a switch.
- General LAN cabling within offices and institutions.

Cross-Wired (Crossover) Cable

- **Definition:** A crossover cable is an Ethernet cable where the transmit and receive wire pairs are swapped at one end. One end follows the **568A** standard and the other end follows **568B**.
- **Purpose:** Used to connect **similar devices** directly (e.g., PC to PC or switch to switch).
- **Wiring:** The positions of the green and orange pairs are swapped. This ensures that the transmit signal from one device matches the receive pin of the other.

Applications:

- Directly connecting two computers without a switch.
- Linking two switches or hubs in the absence of an uplink port.
- Testing and troubleshooting network devices.

Tools for Cable Implementation

1. **Clamping (Crimping) Tool:**
   - Used to attach RJ-45 connectors to the ends of twisted pair cables.
   - Functions include cutting, stripping insulation, and crimping the connector firmly onto the wires.
2. **Cable Tester (Optional):**
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   Verifies the connectivity and correctness of wiring (checks continuity, shorts, and pin alignment).
3. **RJ-45 Connectors:**
   - 8-pin connectors used to terminate twisted pair cables, essential for Ethernet connections.

Practical Implementation Steps

1. Cut the cable to the desired length.
2. Strip about 1–2 cm of the outer insulation carefully, exposing 4 twisted pairs.
3. Untwist the pairs and arrange them in the required wiring order (either 568A or 568B standard).
4. Flatten the wires and trim them to equal length.
5. Insert the wires into an RJ-45 connector, ensuring proper alignment.
6. Place the connector into the crimping tool and press firmly to secure the pins into the cable.
7. Repeat the same process for the other end, following straight-through or crossover configuration.
8. Test the cable with a cable tester to ensure proper functioning.

Importance of Learning Cable Implementation

- Provides hands-on understanding of the physical layer of networking.
- Develops technical skills in handling real-world networking equipment.
- Helps students troubleshoot and build networks effectively, bridging the gap between theory and practice.

LEARNING OUTCOME: The student should have the ability to

LO1.1 To understand the different types of network cables and their applications in computer networks.

LO1.2 To distinguish between straight-through and cross-wired cables based on wiring standards.

LO1.3 To use a clamping tool and RJ-45 connectors for practical cable termination.

LO1.4 To implement and test straight-through and cross-wired cables for establishing physical network connectivity.

COURSE OUTCOME: Upon completion of this course, students will be able to identify various types of network cables and practically implement straight-through and cross-wired cables using clamping tools, thereby gaining hands-on skills in establishing reliable physical layer connectivity in computer networks.

CONCLUSION:

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For Faculty Use

Correction Parameters	Formative Assessment [40%]	Timely completion of Practical [40%]	Attendance / Learning Attitude [20%]	
Marks Obtained				